Where does weight-space programming fix failures?

NVIDIA Cosmos3 Edge Policy (world-action model) and NVIDIA GR00T N1.7 DROID (VLA on a Cosmos-Reason2-2B backbone), with OpenPI π0.5 as the control, on the same RoboLab DROID tasks T1–T8 and a retention task R1.

Method. When a clearer wording of the same goal makes a robot model succeed, copy that wording's internal effect into the weights in closed form, one least-squares solve per layer: no training, and the instruction stays the same.

π0.5, the control (earlier records). Distillation fixed its two language failures: T1 27 → 106 of 128 unseen starts (81 fixed, 2 broke, $p = 7.2 \times 10^{-22}$) and T8 0 → 27 of 64. A random edit of the same size did nothing (9 vs 13 of 64); the T1 edit also lifted T2 and T3 and harmed nothing detectably; no edit helped T4–T7.

NVIDIA models. Cosmos3 Edge Policy: running; GR00T N1.7: running. Same tasks, slots, metrics and statistics; this brief refreshes as runs finish.

What weight-space programming did on each task, on unseen starts

	Cosmos3 Edge Policy world-action model	GR00T N1.7 reasoning-backbone VLA	π0.5 OpenPI VLA, control
T1 Cube then banana	*running*	*running*	fixed: 27 → 106 of 128 distilled edit; 81 / 2, $p = 7.2 \times 10^{-22}$
T2 Pick up the eyeglasses	*running*	*running*	lifted by the T1 edit: 22 → 44 of 64 its own edit: 18 → 27 of 64, not significant
T3 Cube left of the bowl	*running*	*running*	lifted by the T1 edit: 8 → 36 of 64 no own edit submitted
T4 Stack yellow on red	*running*	*running*	no gain lookalike perception; no clear wording to distil
T5 Mustard into the left bin	*running*	*running*	no gain its own edit: 8 → 11 of 64; the grasp slips
T6 Stack three cubes	*running*	*running*	no gain its own edit: 35 → 39 of 64
T7 Pick up the flat box	*running*	*running*	no gain a skill the policy lacks; 0 of 64 unedited
T8 Mustard right of the raisin box, reference-first	*running*	*running*	fixed: 0 → 27 of 64 distilled edit; 27 / 0, $p = 1.5 \times 10^{-8}$
R1 Retention task	*running*	*running*	kept: 32 → 32 of 32 cube into the bowl, T1 edit
Random control, same size	*running*	*running*	9 of 64 T1; unedited 13, edit 50
Where to edit	*running*	*running*	VLM 55, action expert 20, head 0 of 64 T1; unedited 5; only the VLM edit transfers

	Cosmos3 Edge	GR00T N1.7	π0.5
T1 Cube then banana	*running*	*running*	4/32 · 13%
T2 Eyeglasses	*running*	*running*	12/32 · 38%
T3 Cube left of bowl	*running*	*running*	5/32 · 16%
T4 Stack yellow on red	*running*	*running*	13/32 · 41%
T5 Mustard into left b	*running*	*running*	6/32 · 19%
T6 Stack three cubes	*running*	*running*	16/32 · 50%
T7 Flat box	*running*	*running*	0/32 · 0%
T8 Mustard, reference-	*running*	*running*	0/32 · 0%
R1 Retention task	*running*	*running*	32/32 · 100%

Unedited success on each shared task; darker is higher. Each cell gives successes of starts.

	Cosmos3 Edge	GR00T N1.7	π0.5
T1 Cube then banana	*running*	*running*	+79 / 128*
T2 Eyeglasses	*running*	*running*	+22 / 64*
T3 Cube left of bowl	*running*	*running*	+28 / 64*
T4 Stack yellow on red	*running*	*running*	+3 / 32
T5 Mustard into left b	*running*	*running*	+5 / 32
T6 Stack three cubes	*running*	*running*	+5 / 32
T7 Flat box	*running*	*running*	±0 / 32
T8 Mustard, reference-	*running*	*running*	not run
R1 Retention task	*running*	*running*	±0 / 32

gain, $p < 0.05$ · loss, $p < 0.05$ · no detectable change
* $p < 0.05$; outlined: the edit's own task

Each model's main edit, unchanged, on every suite task: edited minus unedited successes on the same starts. Teal: a gain, dark: a loss (exact McNemar $p < 0.05$); grey: no detectable change; the outlined cell is the task the edit was built for.

Live page with every table, figure and record: ali-adeeb.com/weightsmith/cosmos · Paper: ali-adeeb.com/weightsmith/paper.pdf · Code and records: github.com/adeeb10abbas/weightsmith

Draft for discussion. Simulation only (RoboLab, Isaac Sim); one released checkpoint per model; shared slots 30000–30031 (practice) and 96000–96063 (held-out), with identical initial scenes. π0.5 cells are earlier records on their own slots until its shared-slot runs land. Exact two-sided McNemar on fixed vs broke pairs; paired bootstrap 95% intervals.